one-period problems if we can rebalance. For large investors, however, there is a two-, or sometimes three- or four-, stage process where an asset allocation decision (which should be a factor allocation decision) is made at the highest level and then asset managers are chosen who specialize in single asset classes. This two-stage process generates misalignments of interest that are costly to the asset owner.

Static asset class benchmarks introduce further costs in addition to the Irrelevance Result in a multistage delegated setting.[12] Van Binsbergen, Brandt, and Koijen (2008) show that smarter, nonstatic benchmarks can overcome the disadvantages of decentralized investment management. These benchmarks often involve leverage and unusual positions of sector, or subasset class, holdings and would be hard to use in practical settings. However, the general idea remains valid: the static, asset class benchmarks so common in the industry today are bad for asset owners. The industry should move to smarter benchmarks. Factor benchmarks, therefore, have a significant role to play both as a tool for investment performance and as a tool for ameliorating agency issues.

3.4. OPTIMAL CONTRACTS

The Irrelevance Result says that linear contracts do not get the fund manager to act in the interests of the asset owner. So we could move to *nonlinear* performance contracts.

Nonlinear Contracts

The pioneering work of Bhattacharya and Pfleiderer's (1985) paper, which opened up the field of delegated asset management, derived an optimal nonlinear contract. It was quadratic and penalized both negative and positive deviations of the return away from the benchmark. It is natural to expect that compensation should be lower when managers underperform. Quadratic contracts seem unnatural because they also result in lower compensation when the manager has outperformed. And the greater the outperformance, the larger the penalty![13] We don't see quadratic contracts in the real world (at least not yet). But we do see plenty of nonlinear contracts in hedge funds and other alternative asset vehicles. Even if a wrong benchmark is chosen, leading the manager to shirk, having an option-type performance fee will help in motivating the manager to increase her effort.[14] Thus nonlinear contracts should be more widely embraced.

[12] This was first analyzed by Sharpe (1981).

[13] Quadratic contracts penalize excess volatility. In the real world, volatility constraints are imposed using tracking-error constraints. Constraints do play an important role in optimal contracts as I further elaborate in Section 3.4.

[14] This is shown by Li and Tiwari (2009).